

LEZIONE 04-21 PT.2

GOOD NIGHT, SLEEP TIGHT.

A Machine Learning Approach to the study of sleep disorders.

SLEEP: *reversible neuro-behavioural state of relative perceptual disengagement from and unresponsiveness to the environment.*

It's:

- Physiological state
- Recurring
- Alternating.

NREM vs REM Sleep

Non-REM (NREM) sleep: three stages, from light (N1) to deep (N3 or Slow Wave Sleep, SWS)

REM (from rapid-eye-movement) Sleep

PARADOXICAL SLEEP:

- Rapid-eye movements (saccades)
- Skeletal muscle atonia

Our **cortical** activity resembles wake.

It is when most dreams occur:
our mind «moves» but our body «stays still»

Regular sleep cycles positively *impact the quality of life.*

RESTORATIVE FUNCTION

MEMORY CONSOLIDATION

COGNITIVE FUNCTIONING

«BRAIN CLEANING»

BRAIN CLEANING is related with glymphatic system. Has a **pivotal function**, removing metabolism waste products from the Central Nervous System.

The lymphatic activity is higher during **SLOW-WAVE SLEEP**, and *lower during wakefulness*.

Amyloid- β :

if not properly cleared, it aggregates into plaques and contributes to neurodegeneration

An Insight into Sleep Disorders



- SLEEP APNEA: choking of air.
- INSOMNIA: remain asleep.
- NASCOLEPSY & EXCESSIVE SLEEPINESS: suddenness sleep attack.
- HYPERSOMNIA: Excessive need of sleeping.
- RESTLESS LEGS SYNDROME: painful sensation of legs.
- PARASOMNIAS: unusual sleep behaviours.

REM Sleep Behaviour Disorder

REM Sleep Behaviour Disorder (RBD) is a REM Parasomnia

Lack of physiological muscle atonia

The subjects enact their dreams

Ofttimes showing violent behaviour

97%

phenoconversion rate to overt α -synucleinopathy at a **14-year follow-up**

90%

Conversion rate to Parkinson's Disease (PD)

A-synucleinopathy is a family of neurovegetative disorder characterized by an accumulation of a-synuclein in neurons. *Is a protein responsible for neurotransmitter release.*

These disorders cost 3 Billion euros in Italy. 10 People worldwide are affected by Parkinson, of which 40% present secondary REM sleep behaviour Disorder (RBD).

POLYSOMNOGRAPHY is an instrument for **diagnosis** sleep disorders.

Very accurate



Many drawbacks

Impractical

Invasive

Often performed in an unfamiliar setting

Time-consuming (manually scored!)



1)

SCORING.

aim is:

MINIMALLY INVASIVE PSG (Polysomnography) & FAST

Machine learning techniques can do an automatic sleep staging, through two approaches:

- Hand-engineered features
- Row data-driven.

Automatic Sleep Stage Classification

It's based on detection of REM sleep behaviour disorder by automated polysomnography analysis.

Deep Network Approach

It's deep learning in automatic sleep staging with a single channel electroencephalography.

Manual feature extraction approach **VS** deep network approach.

- Feature-based approached are better for interpretability
- Deep networks do not provide explainability, always look for a trade-off.

2)

AUTOMATIC RBD DETECTION

Is faster (but reliable) diagnosis.

This study tries to discover what happens at **cortical level**, investigating the contribution of REM sleep and Slow-wave sleep.

FEATURE EXTRACTION

Time domain
Frequency domain
Time-Frequency domain
Non-Linear domain

3)

OBJECTIVE OUTCOMES

Low cost, continuous monitoring for personalised

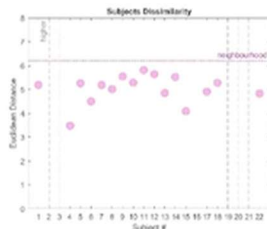
follow-up.

Mapping the Dissociation Index



SIMILARITY MEASURE

Euclidean Distance between
subject and reference



The distance values are
mapped to an index

$$DI = \frac{H_i - \min}{\max - \min}$$

Virtually describing the degree
of dissociation



DISSOCIATION INDEX

Range [0, 1]

Independent and
reproducible metric

It describes the index of REM dissociation. **REM sleep features muscle atonia**. When this atonia does not occur, we say that the subject is in a **REM DISSOCIATUS state**. His brain is in the REM state, but muscle are dissociated than that stage.

Tested on healthy (HS) and RBD subjects

Lower values imply stronger similarity to the
healthy model

Increasing values of Dissociation Index suggest
higher degree of impairment

